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In [38]: import pandas as pd
import numpy as np
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In [39]: from selenium import webdriver
from selenium.webdriver.common.by import By
```

```
In [40]: import math
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```
In [41]: import time
```

```
In [42]: import pickle
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```
In [43]: links = []
total_jobs = []

driver = webdriver.Chrome()
driver.get("https://merojob.com/category")

time.sleep(4)
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
time.sleep(3)
driver.execute_script("window.scrollTo(0, 0)")
time.sleep(3)

for i in range(1,21):
    try:
        link = driver.find_element(By.XPATH, f'/html/body/div/div/main/div/div[2]/d
        links.append(link)
    except Exception as e:
        print(e)

    try:
        total = driver.find_element(By.XPATH, f'/html/body/div/div/main/div/div[2]/
        total_jobs.append(int(total[1:-1]))
    except Exception as e:
        print(e)

driver.quit()

print("!! Link Extracted Successfully !!")
```

!! Link Extracted Successfully !!

```
In [45]: links
```

```
Out[45]: ['https://merojob.com/category/accounting-finance',
          'https://merojob.com/category/architecture-interior-designing',
          'https://merojob.com/category/banking-insurance-financial-services',
          'https://merojob.com/category/commercial-logistics-supply-chain',
          'https://merojob.com/category/construction-engineering-architects',
          'https://merojob.com/category/creative-graphics-designing',
          'https://merojob.com/category/education-counseling-career-counseling',
          'https://merojob.com/category/general-mgmt-administration-operations',
          'https://merojob.com/category/healthcare-pharma-biotech-medical-rd',
          'https://merojob.com/category/hospitality',
          'https://merojob.com/category/human-resource-org-development',
          'https://merojob.com/category/it-telecommunication',
          'https://merojob.com/category/legal-services',
          'https://merojob.com/category/marketing-advertising-customer-service',
          'https://merojob.com/category/ngo-ingo-social-work',
          'https://merojob.com/category/production-maintenance-quality',
          'https://merojob.com/category/research-and-development',
          'https://merojob.com/category/sales-public-relations',
          'https://merojob.com/category/secretarial-front-office-data-entry',
          'https://merojob.com/category/teaching-education']
```

```
In [46]: total_jobs
```

```
Out[46]: [58, 7, 1, 4, 25, 9, 44, 304, 13, 17, 13, 37, 4, 94, 7, 5, 22, 127, 24, 59]
```

```
In [47]: job_dict = dict(zip(links, total_jobs))
```

```
In [7]: with open(r'C:\Users\hp\Desktop\Jupyter Notebook\WebScraping\links.pkl', 'wb') as f:
        pickle.dump(links, file)
```

```
In [48]: with open(r'C:\Users\hp\Desktop\Jupyter Notebook\WebScraping\links.pkl', 'rb') as f:
        loaded_data = pickle.load(file)
```

```
In [49]: loaded_data
```

```
Out[49]: ['https://merojob.com/category/accounting-finance',
'https://merojob.com/category/architecture-interior-designing',
'https://merojob.com/category/banking-insurance-financial-services',
'https://merojob.com/category/commercial-logistics-supply-chain',
'https://merojob.com/category/construction-engineering-architects',
'https://merojob.com/category/creative-graphics-designing',
'https://merojob.com/category/education-counseling-career-counseling',
'https://merojob.com/category/general-mgmt-administration-operations',
'https://merojob.com/category/healthcare-pharma-biotech-medical-rd',
'https://merojob.com/category/hospitality',
'https://merojob.com/category/human-resource-org-development',
'https://merojob.com/category/it-telecommunication',
'https://merojob.com/category/legal-services',
'https://merojob.com/category/marketing-advertising-customer-service',
'https://merojob.com/category/ngo-ingo-social-work',
'https://merojob.com/category/production-maintenance-quality',
'https://merojob.com/category/research-and-development',
'https://merojob.com/category/sales-public-relations',
'https://merojob.com/category/secretarial-front-office-data-entry',
'https://merojob.com/category/teaching-education']
```

```
In [50]: import time
import numpy as np
import pandas as pd
from selenium import webdriver
from selenium.webdriver.common.by import By
import math

def extract_data(url, total):

    driver = webdriver.Chrome()

    job_title = []
    company = []
    level = []
    employment_type = []
    location = []
    experience_required = []
    salary = []
    skills_required = []
    deadline = []

    category = url.split('/')[ -1]
    limit = 20
    total_pages = math.ceil(total / limit)

    heading = category.replace("-", " ").title()
    print(heading)

    try:
        for page in range(1, total_pages+1):
            links = url + f'?limit={limit}&offset={page}'

            driver.get(links)

            time.sleep(5)
```

```
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
time.sleep(3)
driver.execute_script("window.scrollTo(0, 0)")
time.sleep(2)

for i in range(1, limit+1):

    # Job Title
    try:
        t = driver.find_element(By.XPATH, f"/html/body/div/div/main/div
        job_title.append(t)
    except:
        job_title.append(np.nan)

    # Company
    try:
        c = driver.find_element(By.XPATH, f"/html/body/div/div/main/div
        company.append(c)
    except:
        company.append(np.nan)

    # Level
    try:
        l = driver.find_element(By.XPATH, f"/html/body/div/div/main/div
        level.append(l)
    except:
        level.append(np.nan)

    # Employment Type
    try:
        et = driver.find_element(By.XPATH, f"/html/body/div/div/main/di
        employment_type.append(et)
    except:
        employment_type.append(np.nan)

    # Location
    try:
        loc = driver.find_element(By.XPATH, f"/html/body/div/div/main/d
        location.append(loc)
    except:
        location.append(np.nan)

    # Experience Required
    try:
        exp = driver.find_element(By.XPATH, f"/html/body/div/div/main/d
        experience_required.append(exp)
    except:
        experience_required.append(np.nan)

    # Salary
    try:
        sal = driver.find_element(By.XPATH, f"/html/body/div/div/main/d
        salary.append(sal)
    except:
        salary.append(np.nan)
```

```

        # Skills Required
        try:
            skl = driver.find_element(By.XPATH, f"/html/body/div/div/main/d
            skills_required.append(skl)
        except:
            skills_required.append(np.nan)

        # Deadline
        try:
            dl = driver.find_element(By.XPATH, f"/html/body/div/div/main/di
            deadline.append(dl)
        except:
            deadline.append(np.nan)

    print(f"--> {page} page extracted successfully.")

driver.quit()

# Create dictionary using ALL lists
data_dict = {
    "job_title": job_title,
    "company": company,
    "level": level,
    "employment_type": employment_type,
    "location": location,
    "experience_required": experience_required,
    "salary": salary,
    "skills_required": skills_required,
    "deadline": deadline
}

df_job = pd.DataFrame(data_dict)

df_job = df_job.dropna(how='all')

df_job.to_csv(f"C:\\Users\\hp\\Desktop\\Jupyter Notebook\\WebScraping\\Mero

print("Data Extraction Completed.\n")
return df_job

except Exception as e:
    print("Error:", e)

```

In []:

In [51]: final = []

```

for link, total_job in zip(links, total_jobs):
    final.append(extract_data(link, total_job))

```

Accounting Finance

--> 1 page extracted successfully.
--> 2 page extracted successfully.
--> 3 page extracted successfully.
Data Extraction Completed.

Architecture Interior Designing

--> 1 page extracted successfully.
Data Extraction Completed.

Banking Insurance Financial Services

--> 1 page extracted successfully.
Data Extraction Completed.

Commercial Logistics Supply Chain

--> 1 page extracted successfully.
Data Extraction Completed.

Construction Engineering Architects

--> 1 page extracted successfully.
--> 2 page extracted successfully.
Data Extraction Completed.

Creative Graphics Designing

--> 1 page extracted successfully.
Data Extraction Completed.

Education Counseling Career Counseling

--> 1 page extracted successfully.
--> 2 page extracted successfully.
--> 3 page extracted successfully.
Data Extraction Completed.

General Mgmt Administration Operations

--> 1 page extracted successfully.
--> 2 page extracted successfully.
--> 3 page extracted successfully.
--> 4 page extracted successfully.
--> 5 page extracted successfully.
--> 6 page extracted successfully.
--> 7 page extracted successfully.
--> 8 page extracted successfully.
--> 9 page extracted successfully.
--> 10 page extracted successfully.
--> 11 page extracted successfully.
--> 12 page extracted successfully.
--> 13 page extracted successfully.
--> 14 page extracted successfully.
--> 15 page extracted successfully.
--> 16 page extracted successfully.
Data Extraction Completed.

Healthcare Pharma Biotech Medical Rd

--> 1 page extracted successfully.
Data Extraction Completed.

Hospitality

--> 1 page extracted successfully.
Data Extraction Completed.

Human Resource Org Development

--> 1 page extracted successfully.
Data Extraction Completed.

It Telecommunication

--> 1 page extracted successfully.
--> 2 page extracted successfully.
Data Extraction Completed.

Legal Services

--> 1 page extracted successfully.
Data Extraction Completed.

Marketing Advertising Customer Service

--> 1 page extracted successfully.
--> 2 page extracted successfully.
--> 3 page extracted successfully.
--> 4 page extracted successfully.
--> 5 page extracted successfully.
Data Extraction Completed.

Ngo Ingo Social Work

--> 1 page extracted successfully.
Data Extraction Completed.

Production Maintenance Quality

--> 1 page extracted successfully.
Data Extraction Completed.

Research And Development

--> 1 page extracted successfully.
--> 2 page extracted successfully.
Data Extraction Completed.

Sales Public Relations

--> 1 page extracted successfully.
--> 2 page extracted successfully.
--> 3 page extracted successfully.
--> 4 page extracted successfully.
--> 5 page extracted successfully.
--> 6 page extracted successfully.
--> 7 page extracted successfully.
Data Extraction Completed.

Secretarial Front Office Data Entry

--> 1 page extracted successfully.
--> 2 page extracted successfully.
Data Extraction Completed.

Teaching Education

--> 1 page extracted successfully.
--> 2 page extracted successfully.

--> 3 page extracted successfully.
Data Extraction Completed.

In [53]: final[5]

Out[53]:

	job_title	company	level	employment_type	location	experience_requirement
1	Marketing Manager	Kama Jewels	Senior Level	Full Time	Kalanki, Kathmandu	More than 5 years
2	Content Creator	The British College	Mid Level	Full Time	5th Floor, Trade Tower, Thapathali, Kathmandu	More than 2 years
3	Graphic Designer	One Pasal	Mid Level	Full Time	Naxal	More than 1 year
4	Graphic Designer	K.D. Shoes Industries	Mid Level	Full Time	Kathmandu	More than 2 years
5	AI and Video Creator	NaN	Mid Level	Full Time	Kathmandu	More than 1 year
6	Social Media Coordinator	Machhapuchchhre World School	Mid Level	Full Time	Corporate Office (serving all organizations un...	More than 1 year
7	Junior Graphic Designer / Video Editor	Mad Ads Agency	Mid Level	Full Time	Kathmandu	More than 1 year



In []: